Úloha 2 – Posilovna



	60 min
Téma	Cvičení, Vzpírání, Pohon, Proměnné, Opakování, Stopky, (Převody)

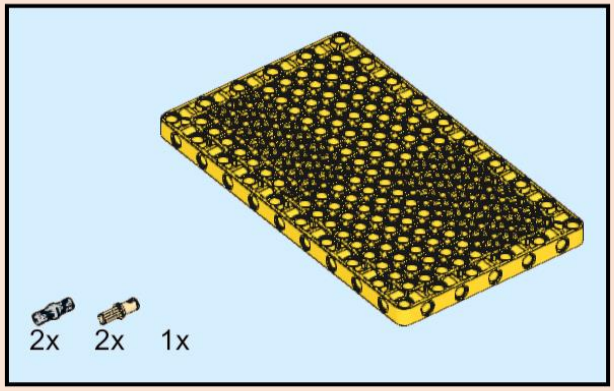


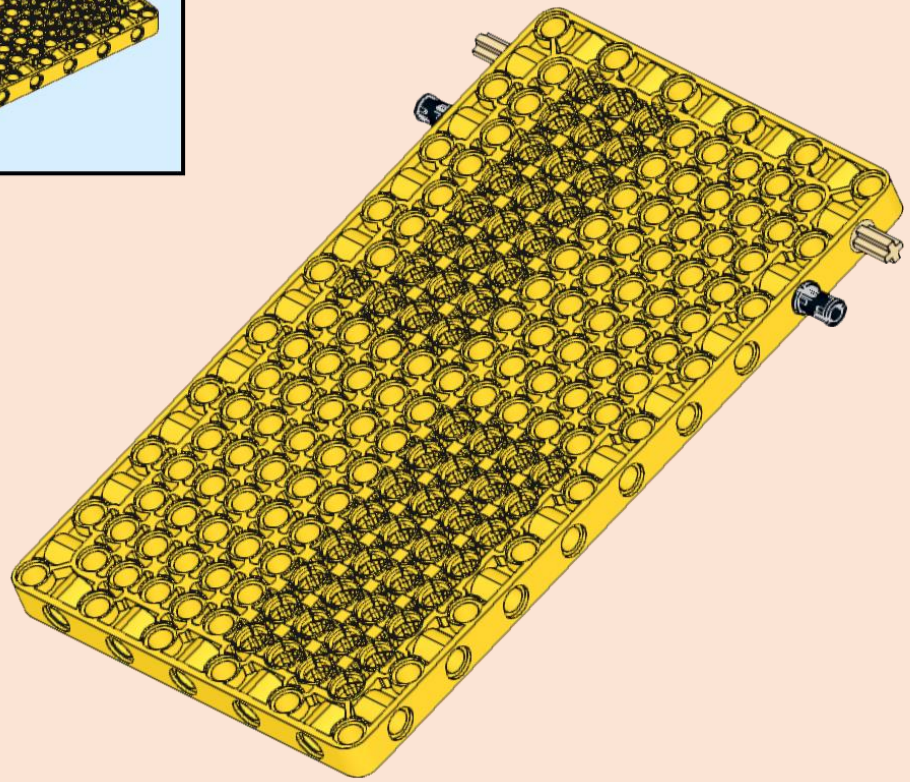
Co sestavit a připravit:

- ☐ Sestroj posilovací lavici i s fitnesákem podle návodu.
- ☐ Vymysli, jak pomocí motoru rozpohybovat činku nahoru a dolu.

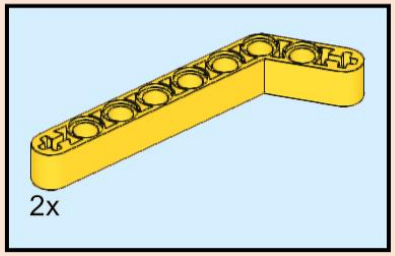


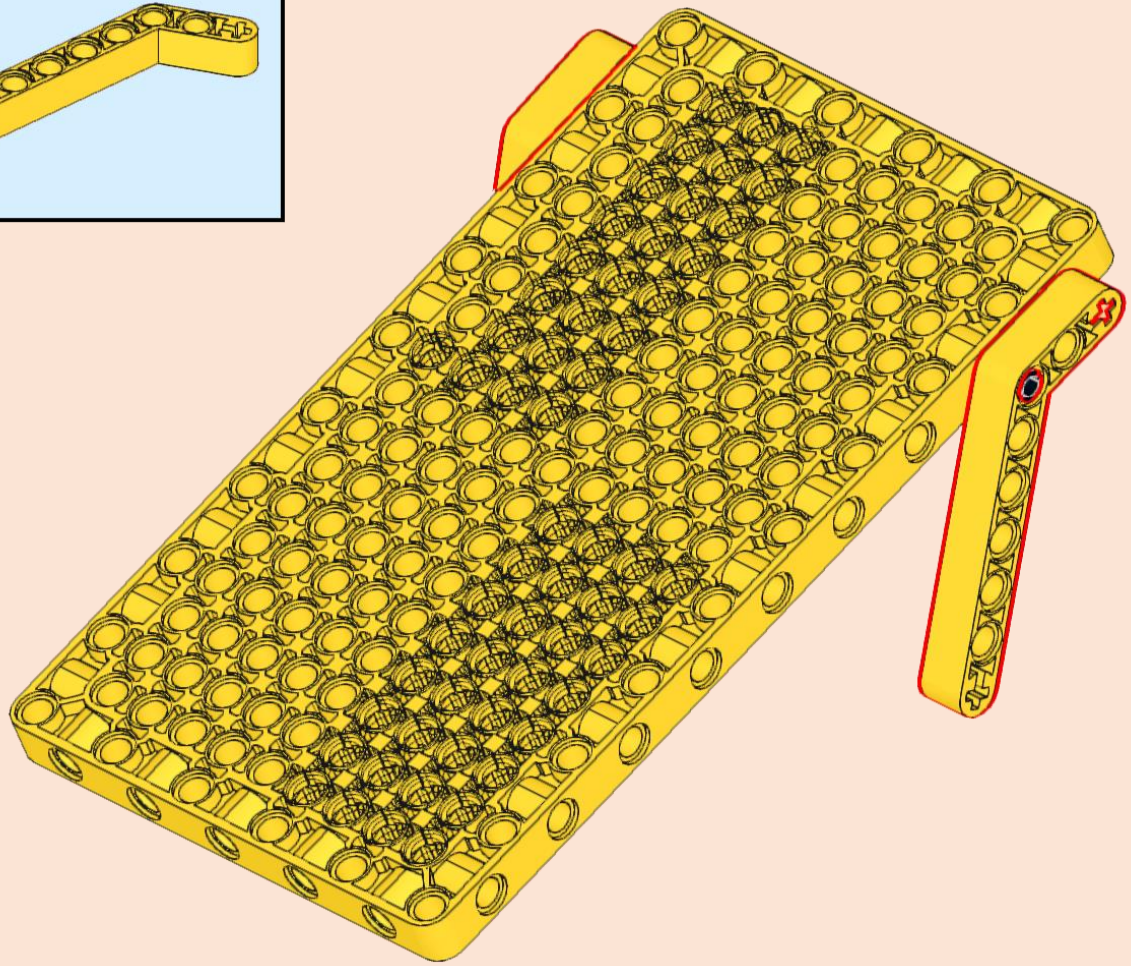
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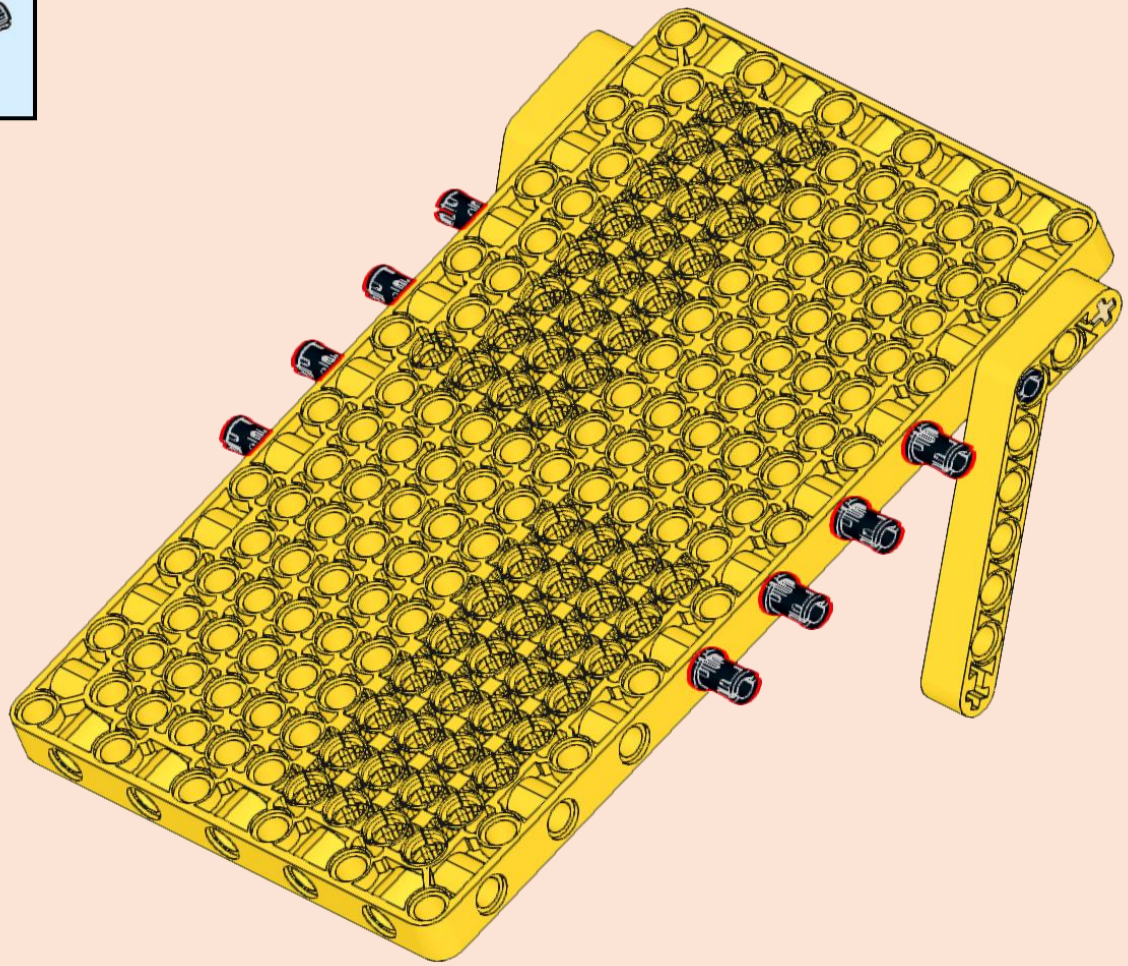
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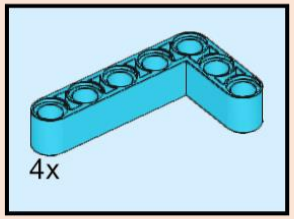


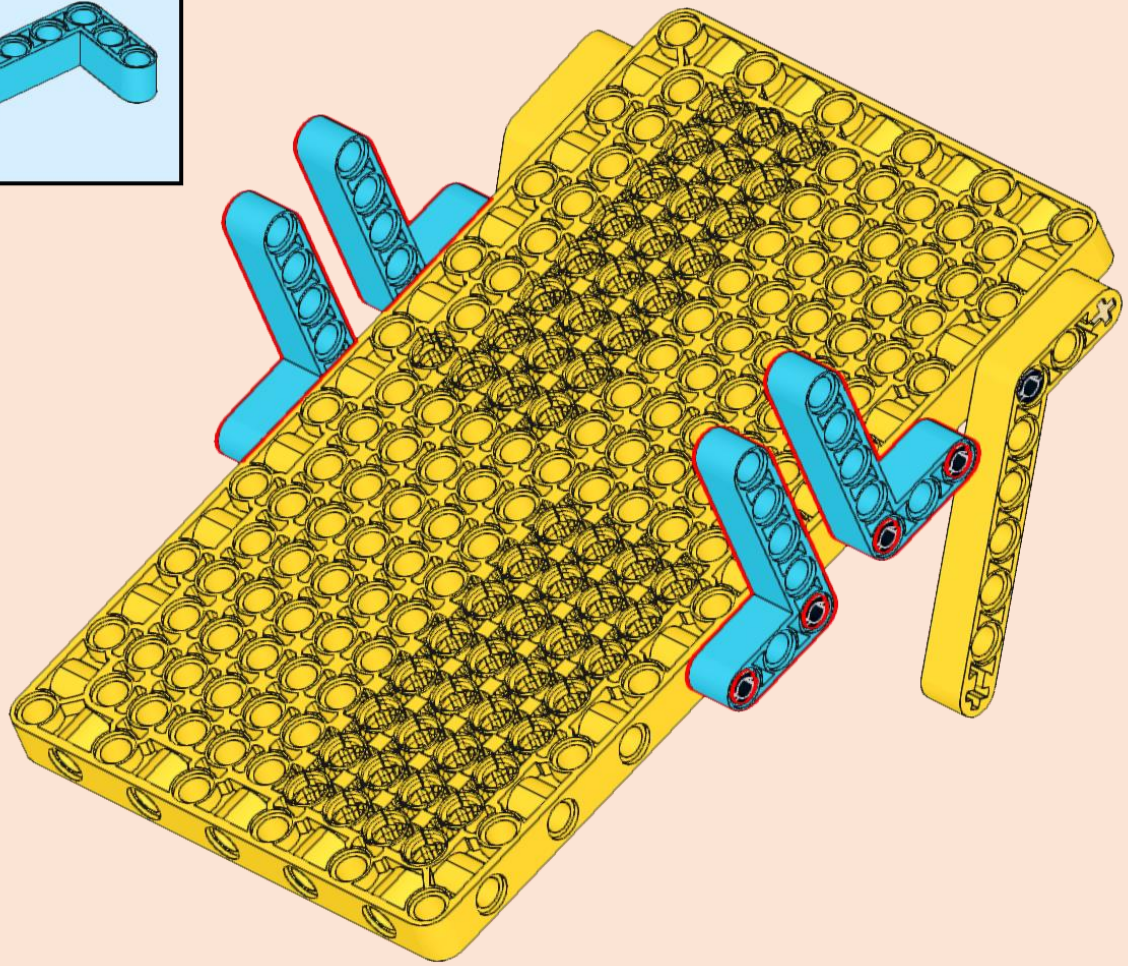
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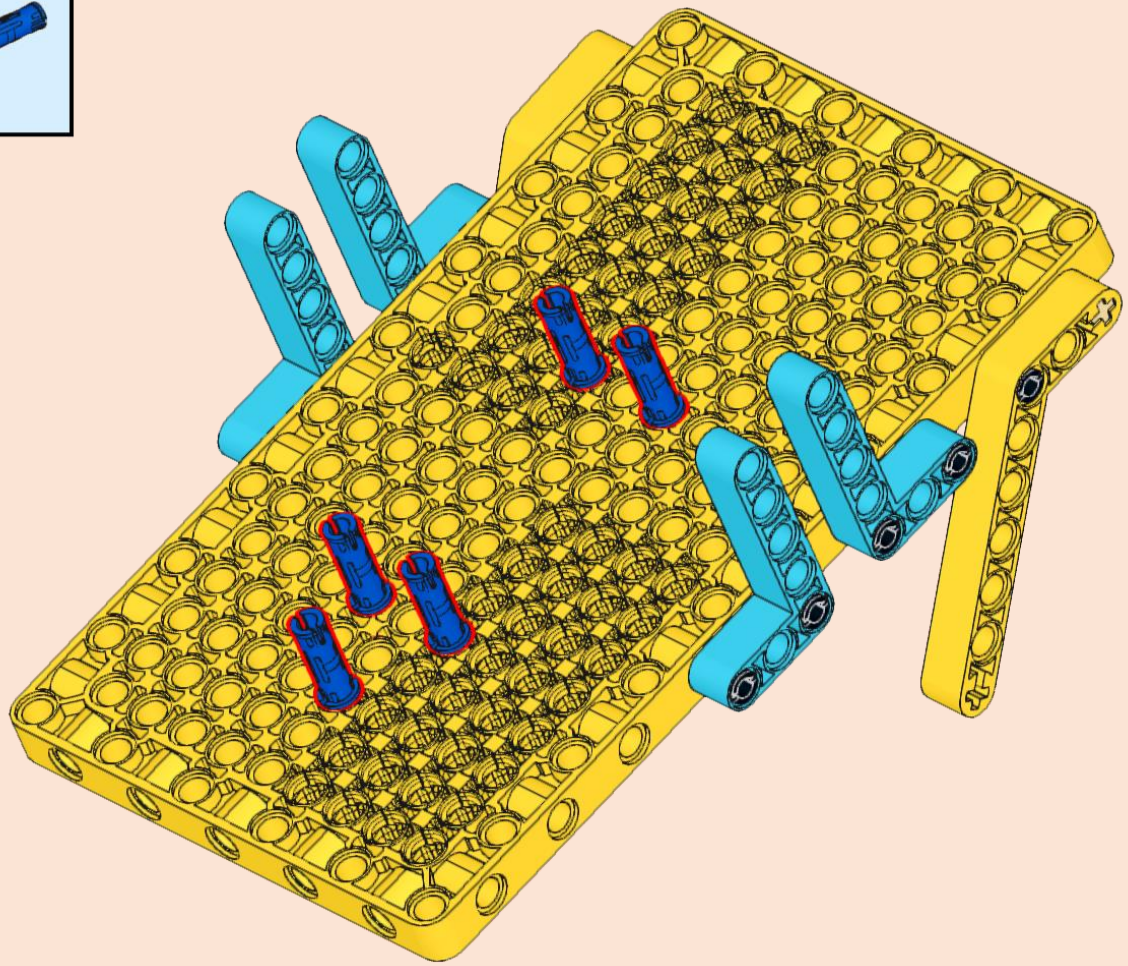
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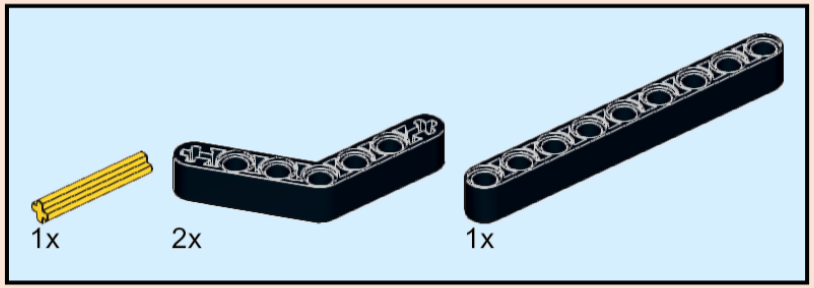


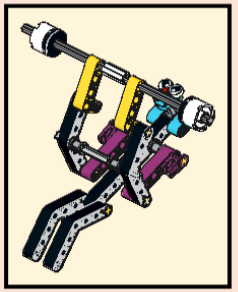
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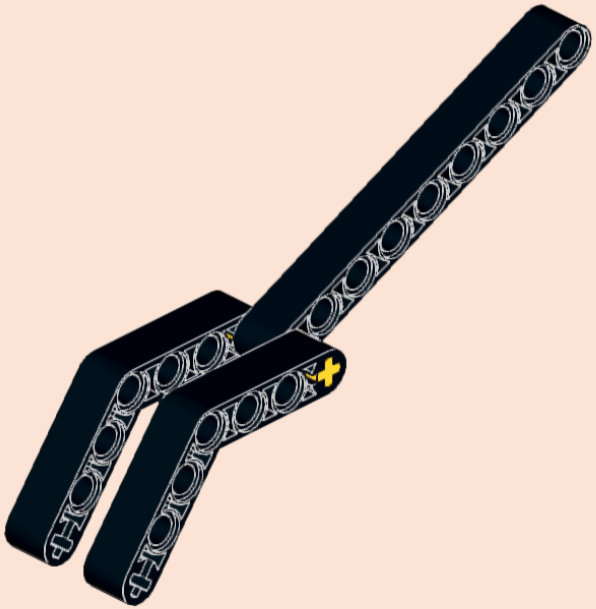




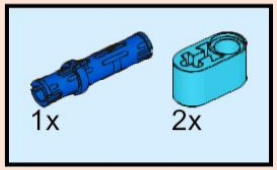
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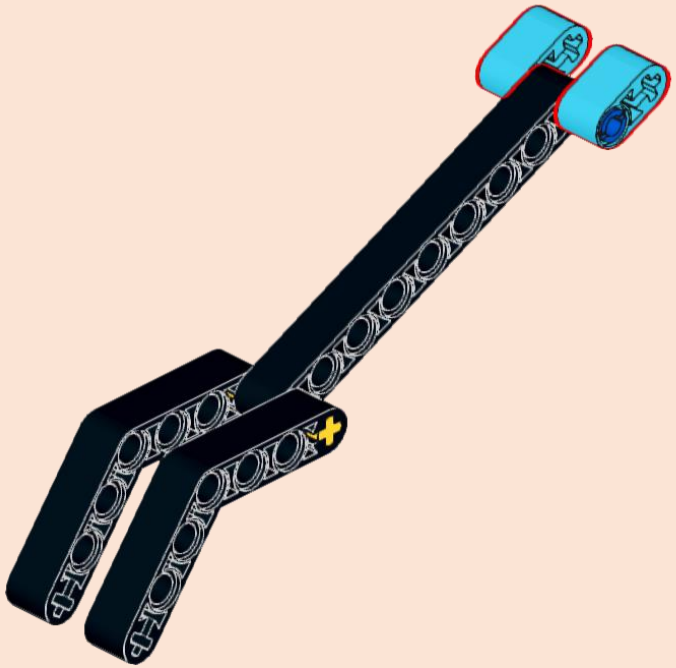




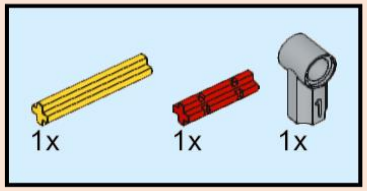


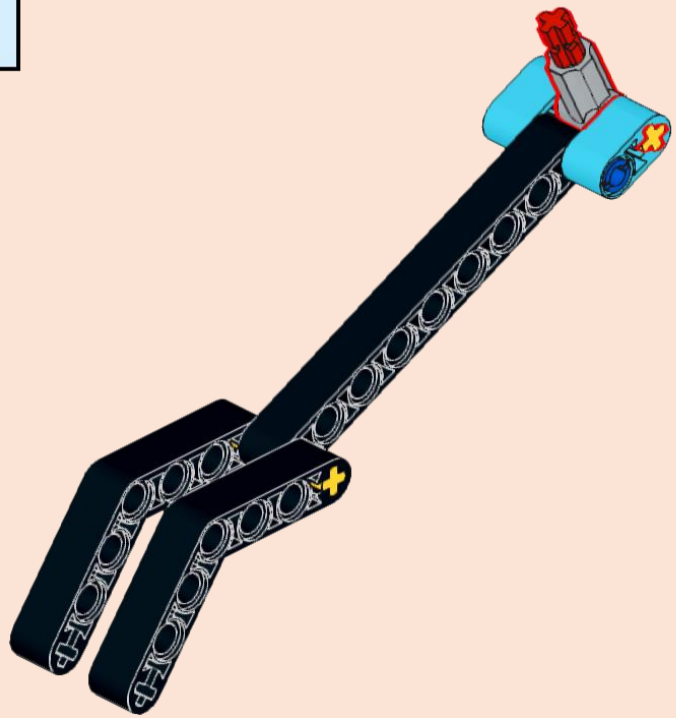
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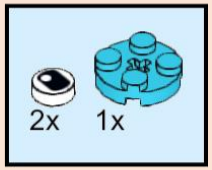


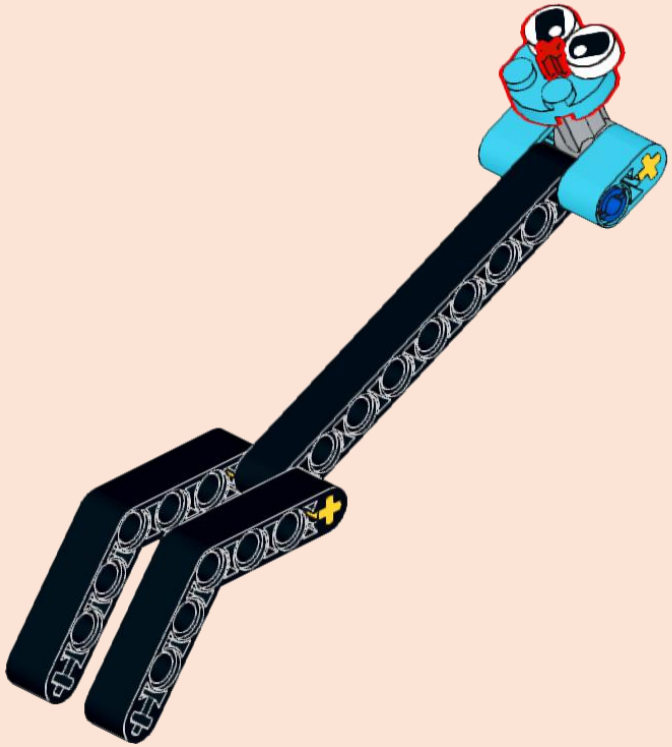
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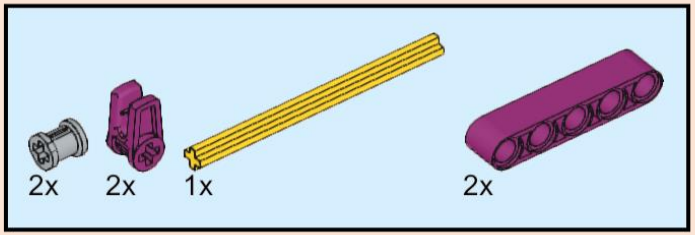


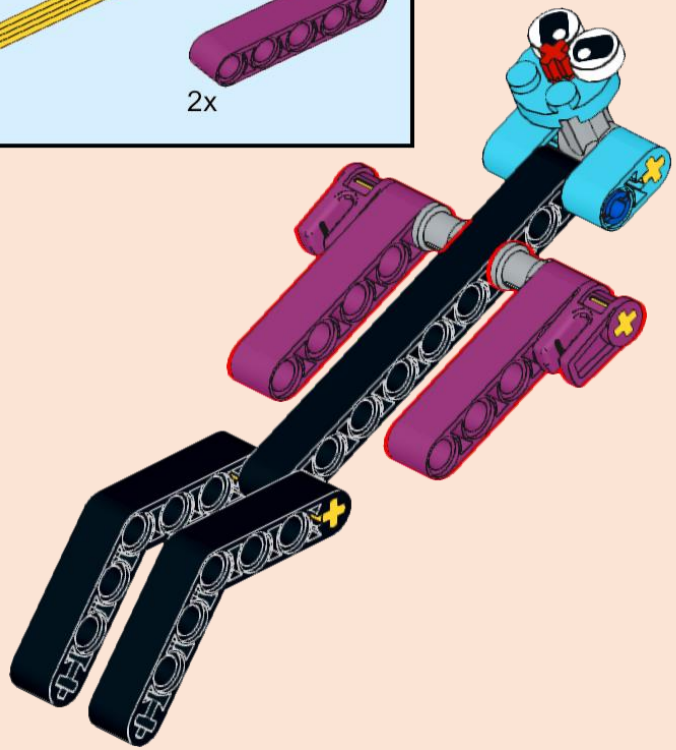
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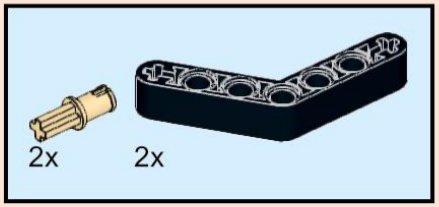


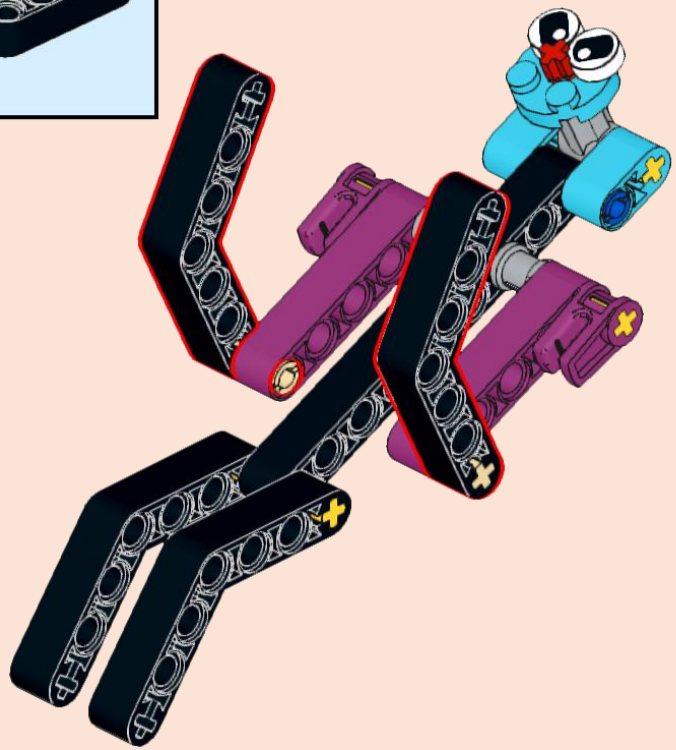
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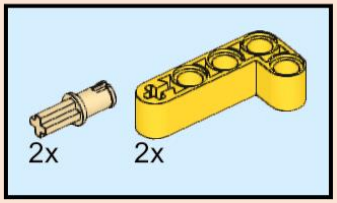


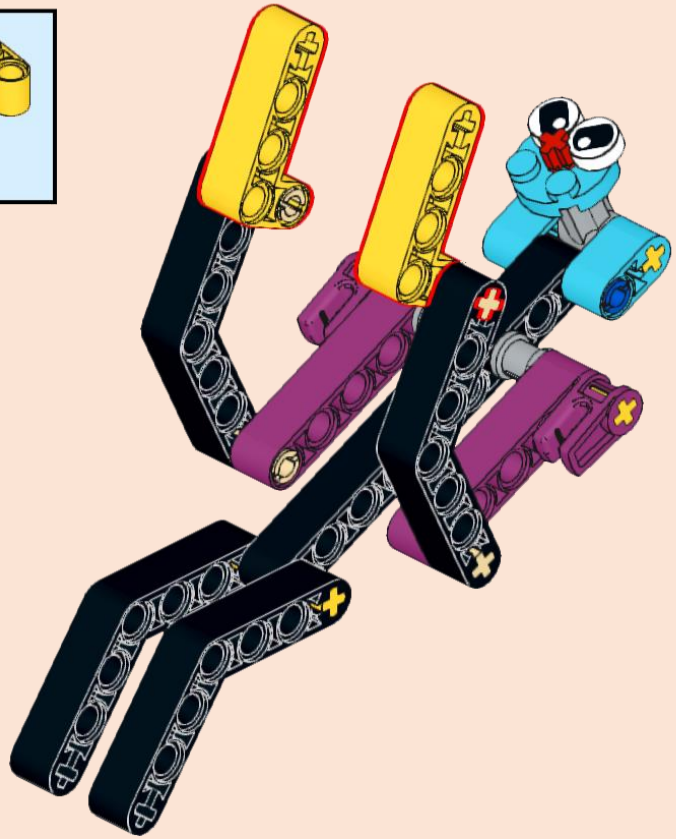
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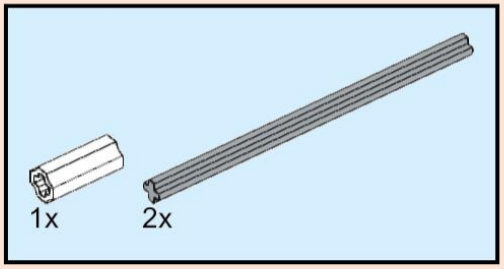


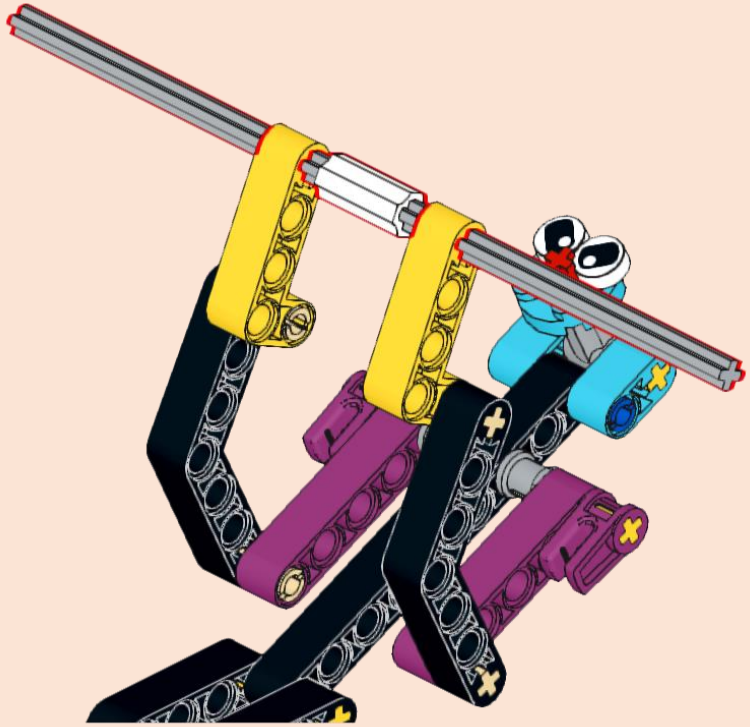
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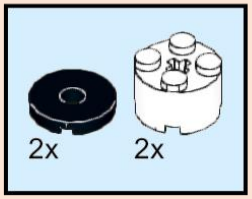


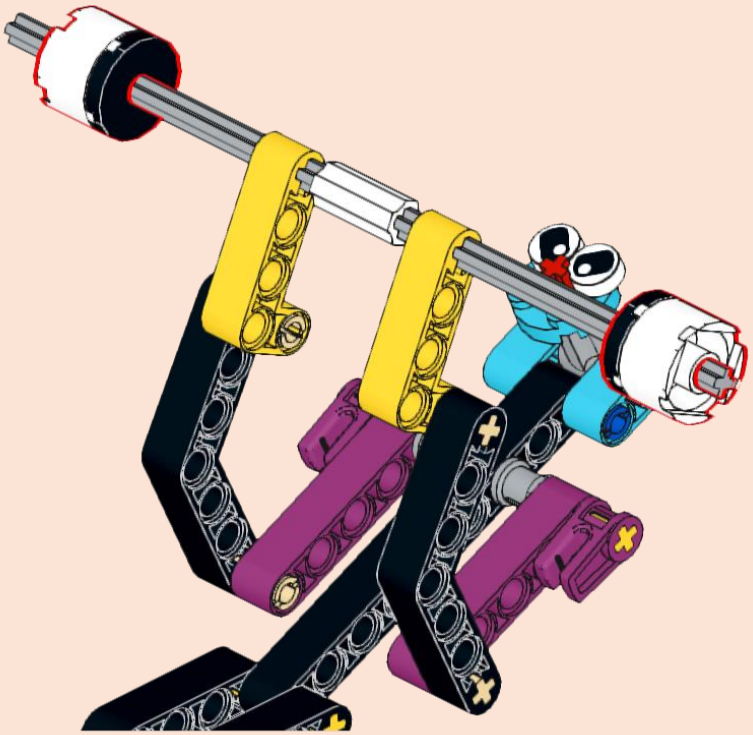
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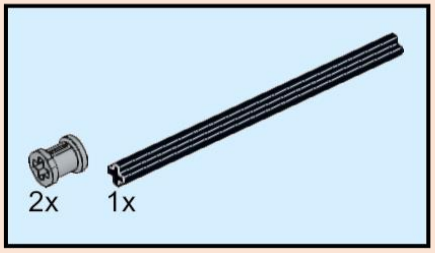


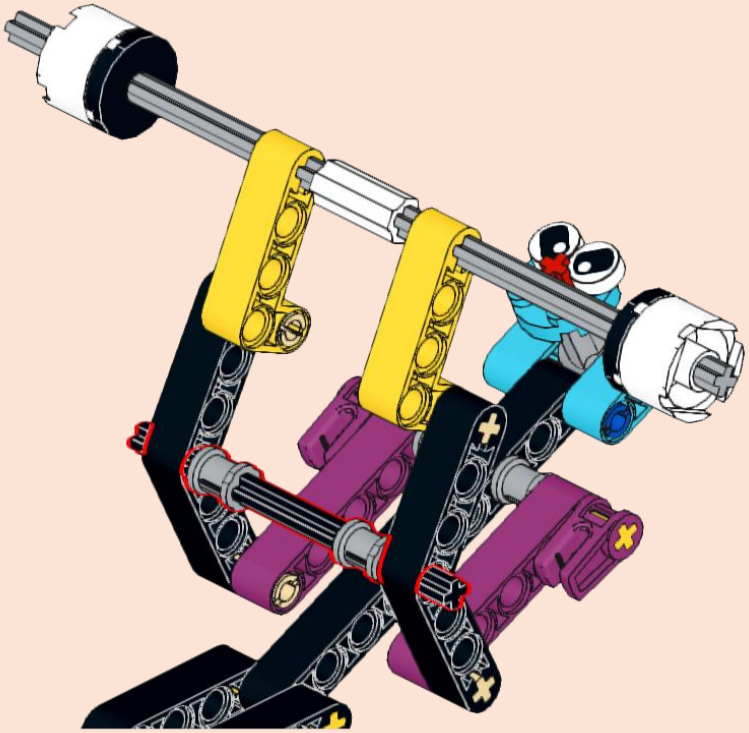
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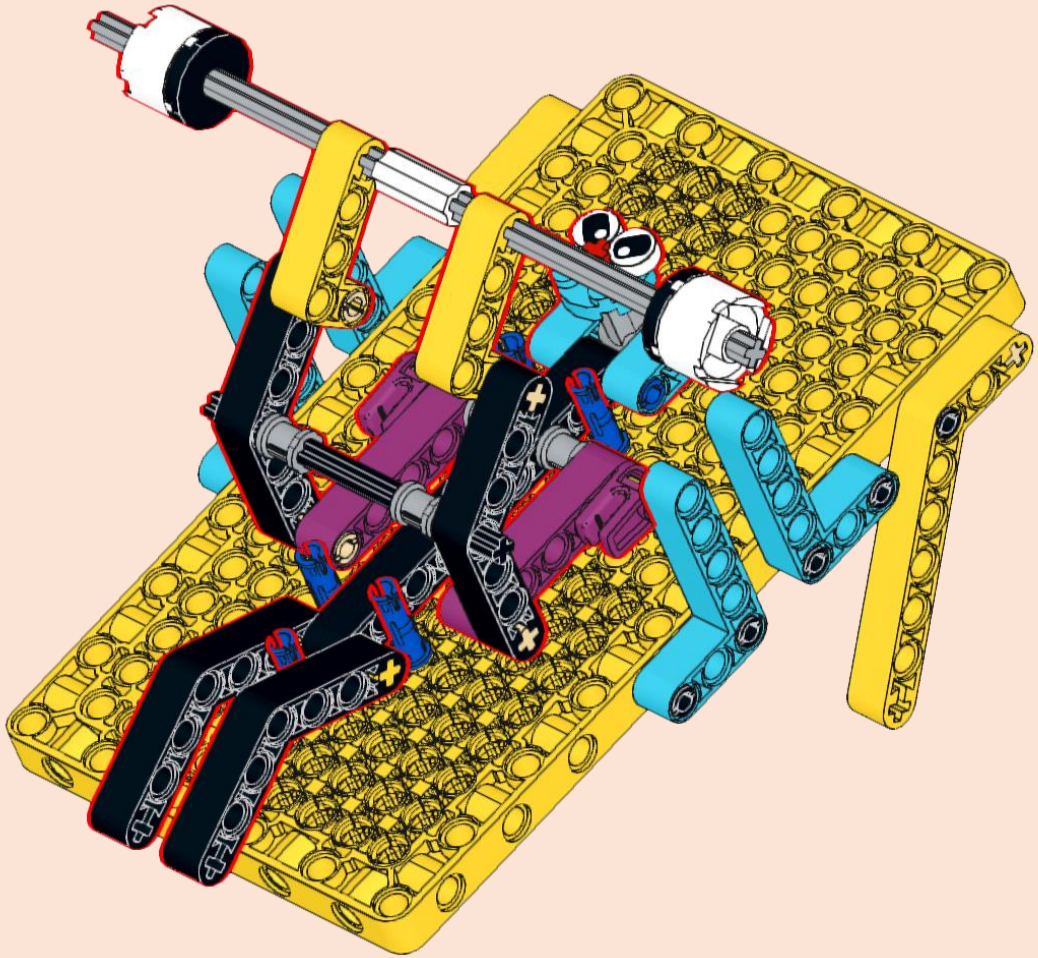


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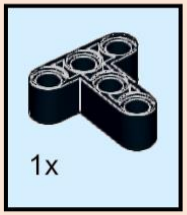


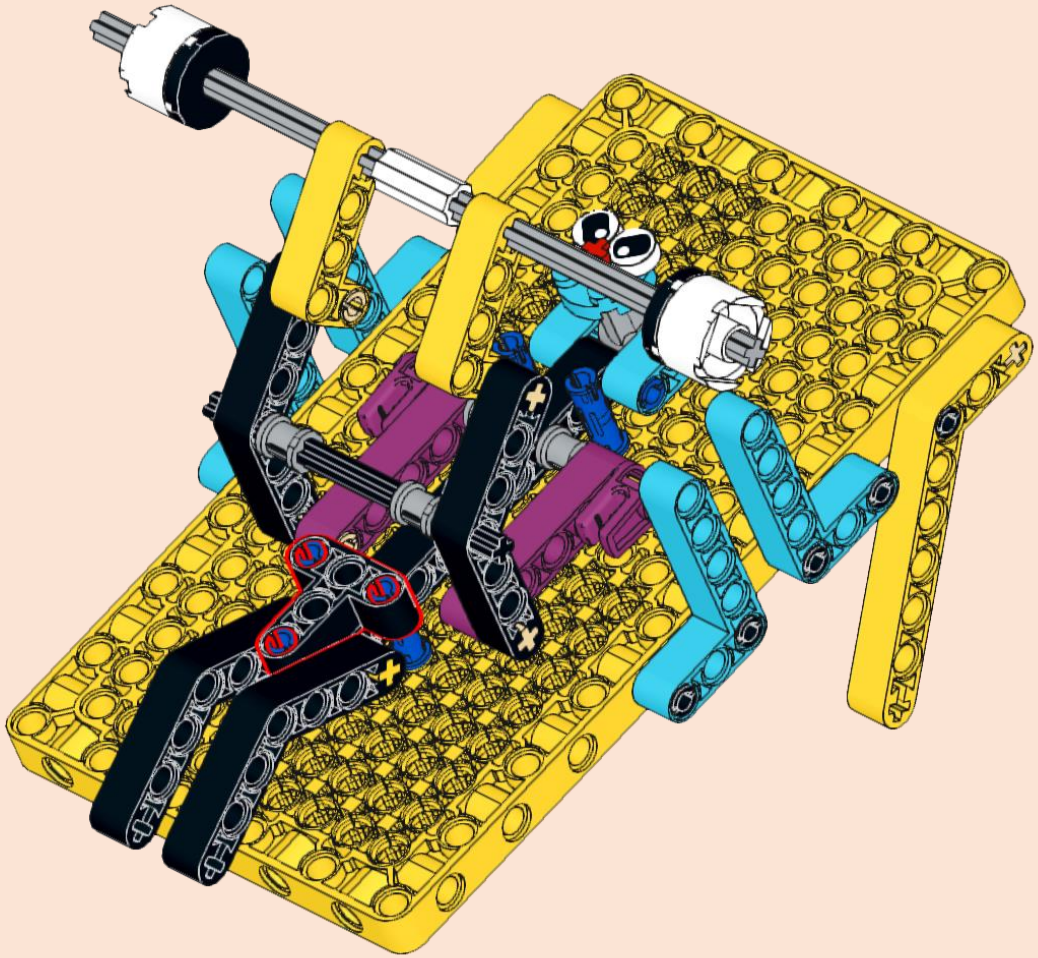


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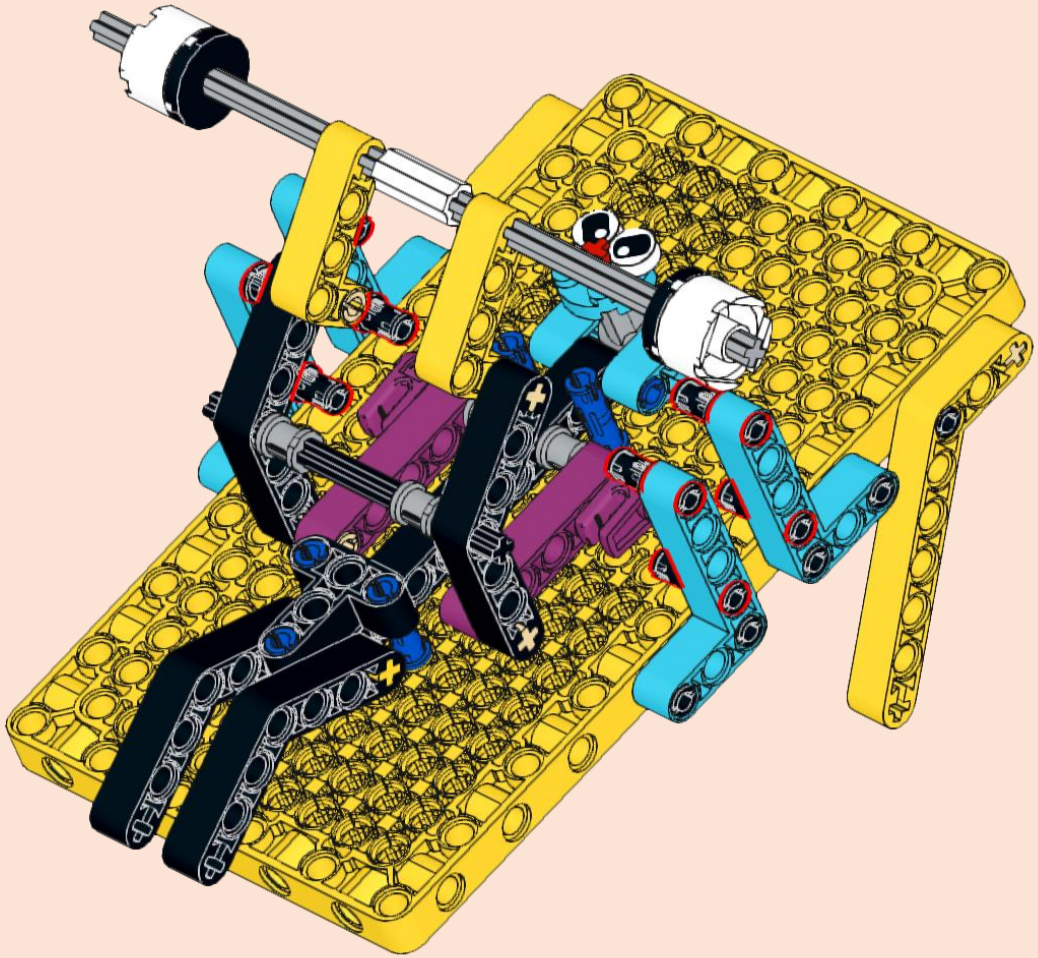
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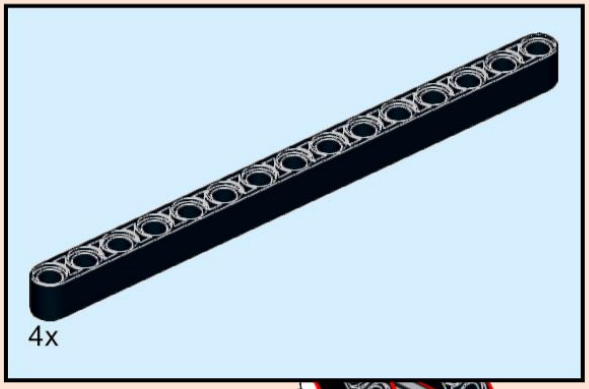


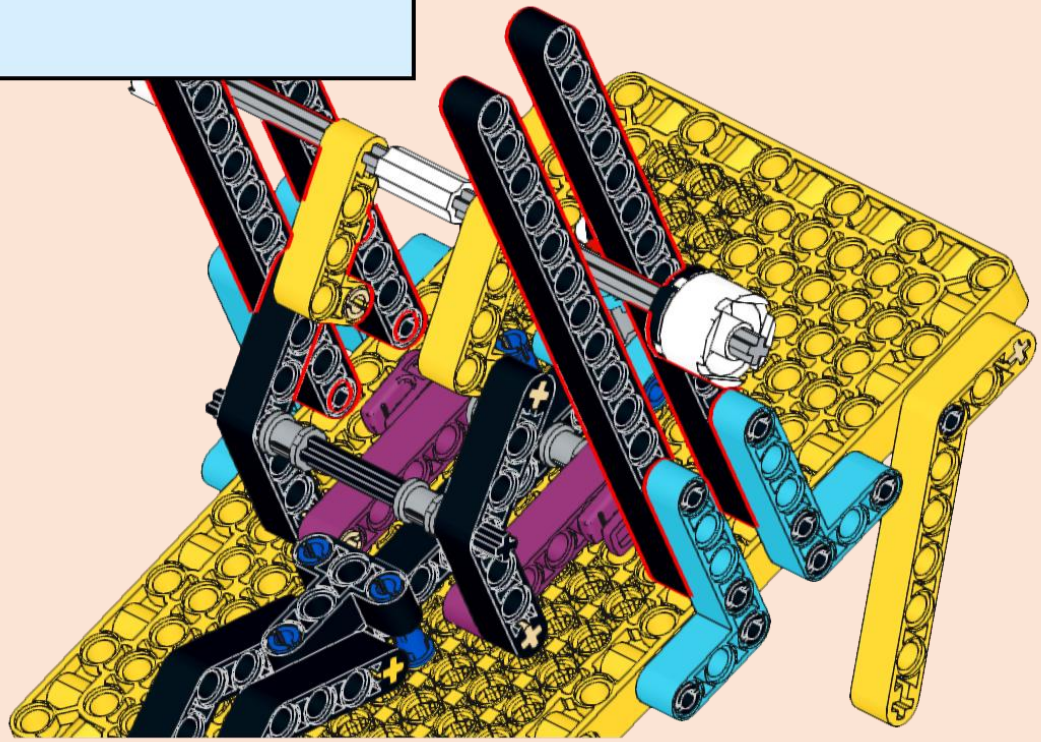
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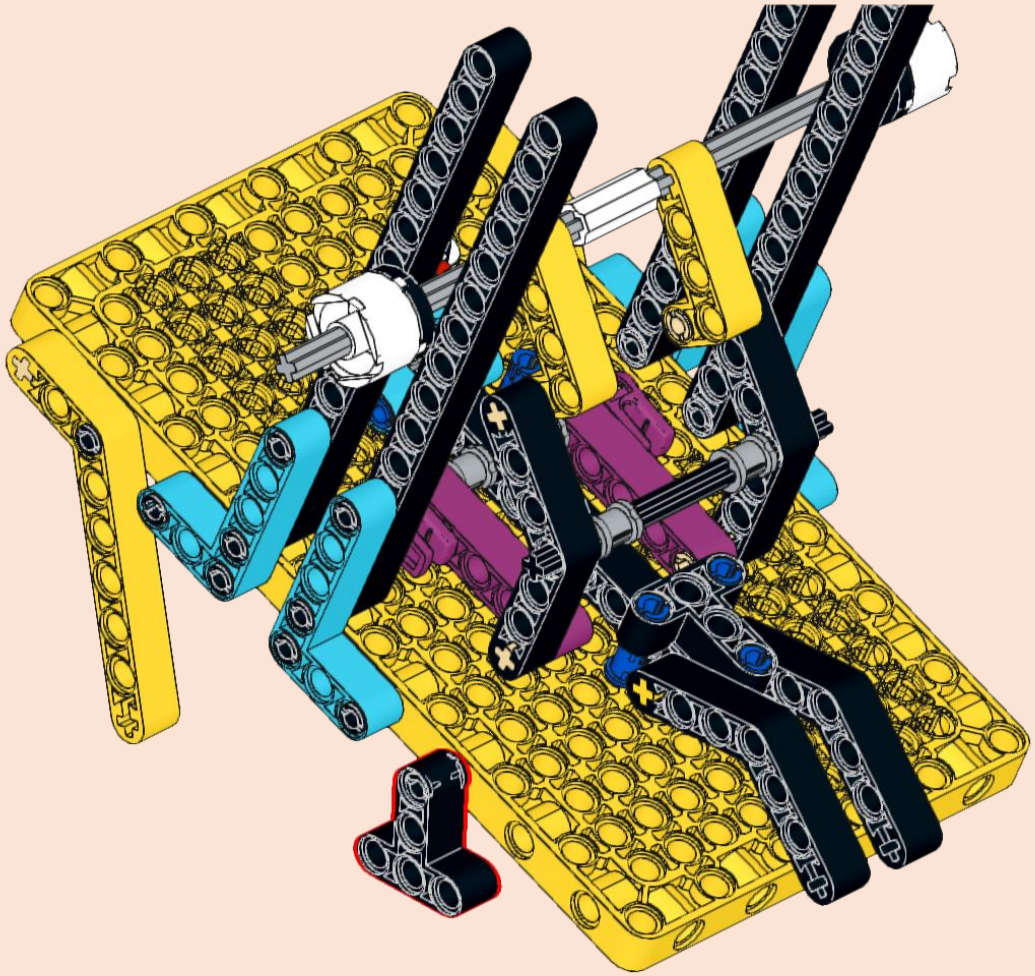


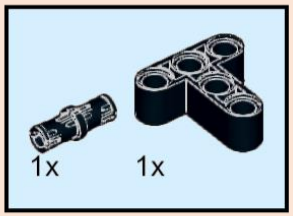
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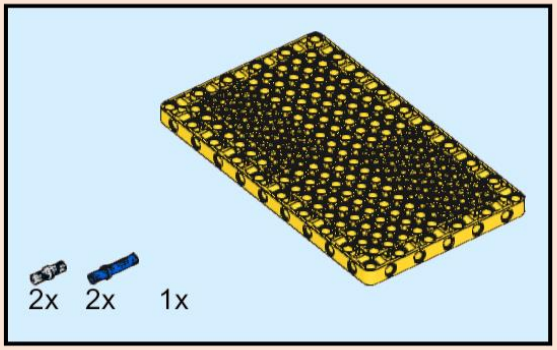


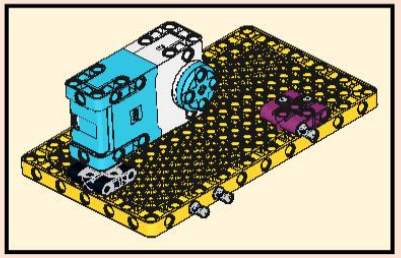
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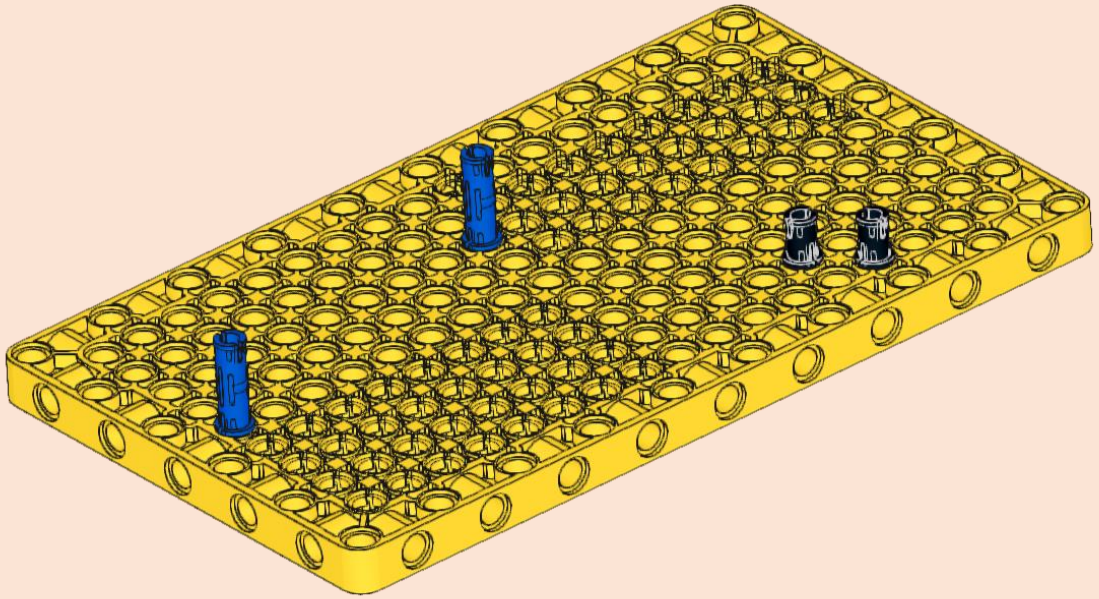




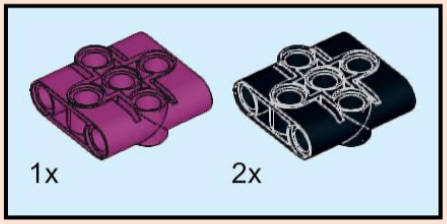
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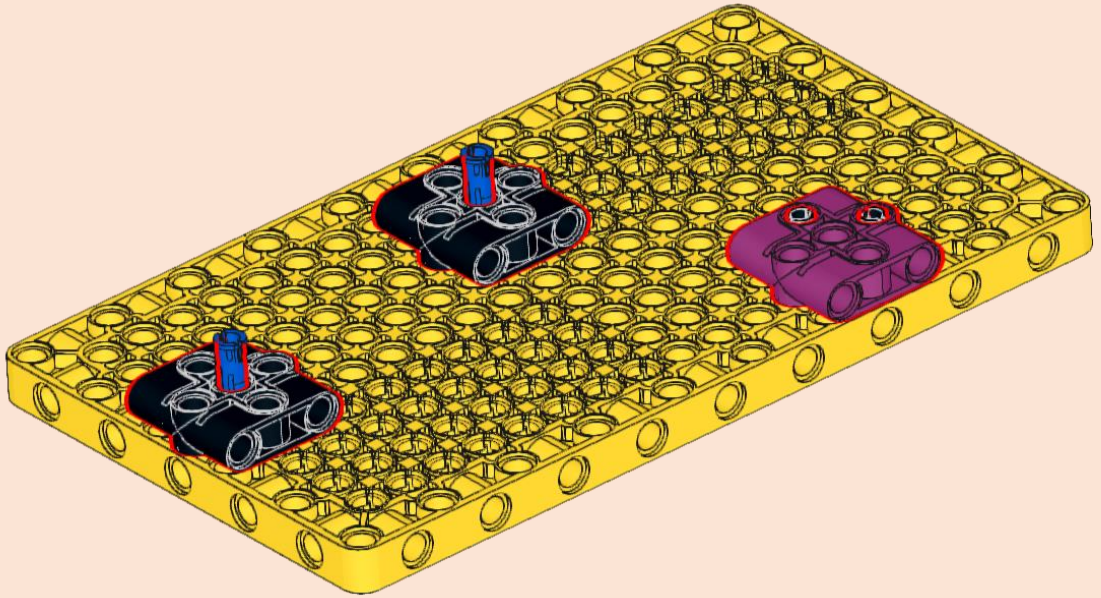




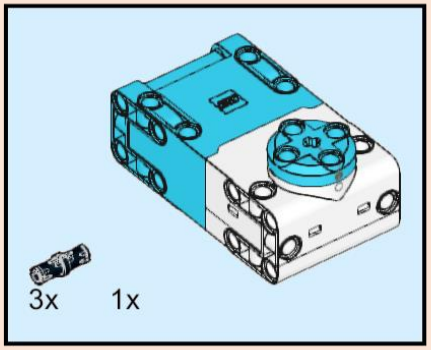


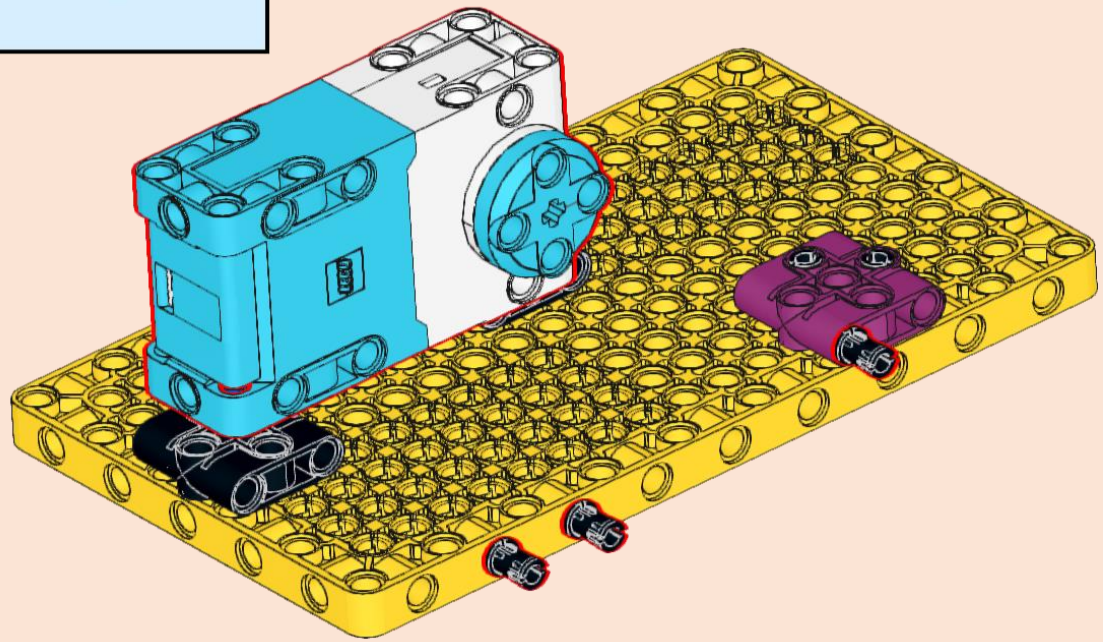
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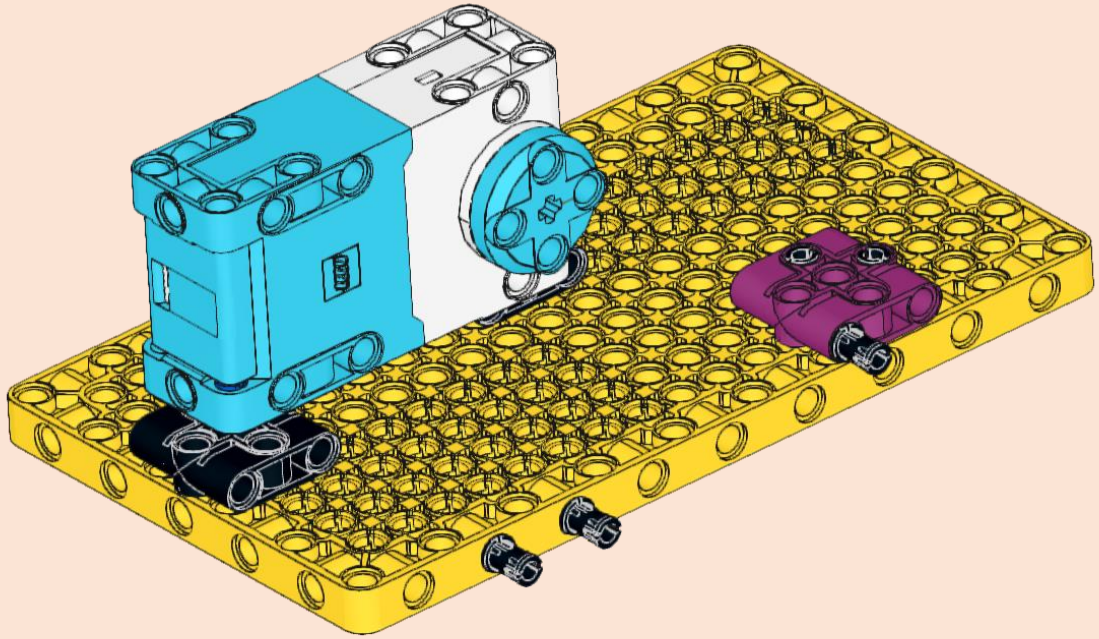


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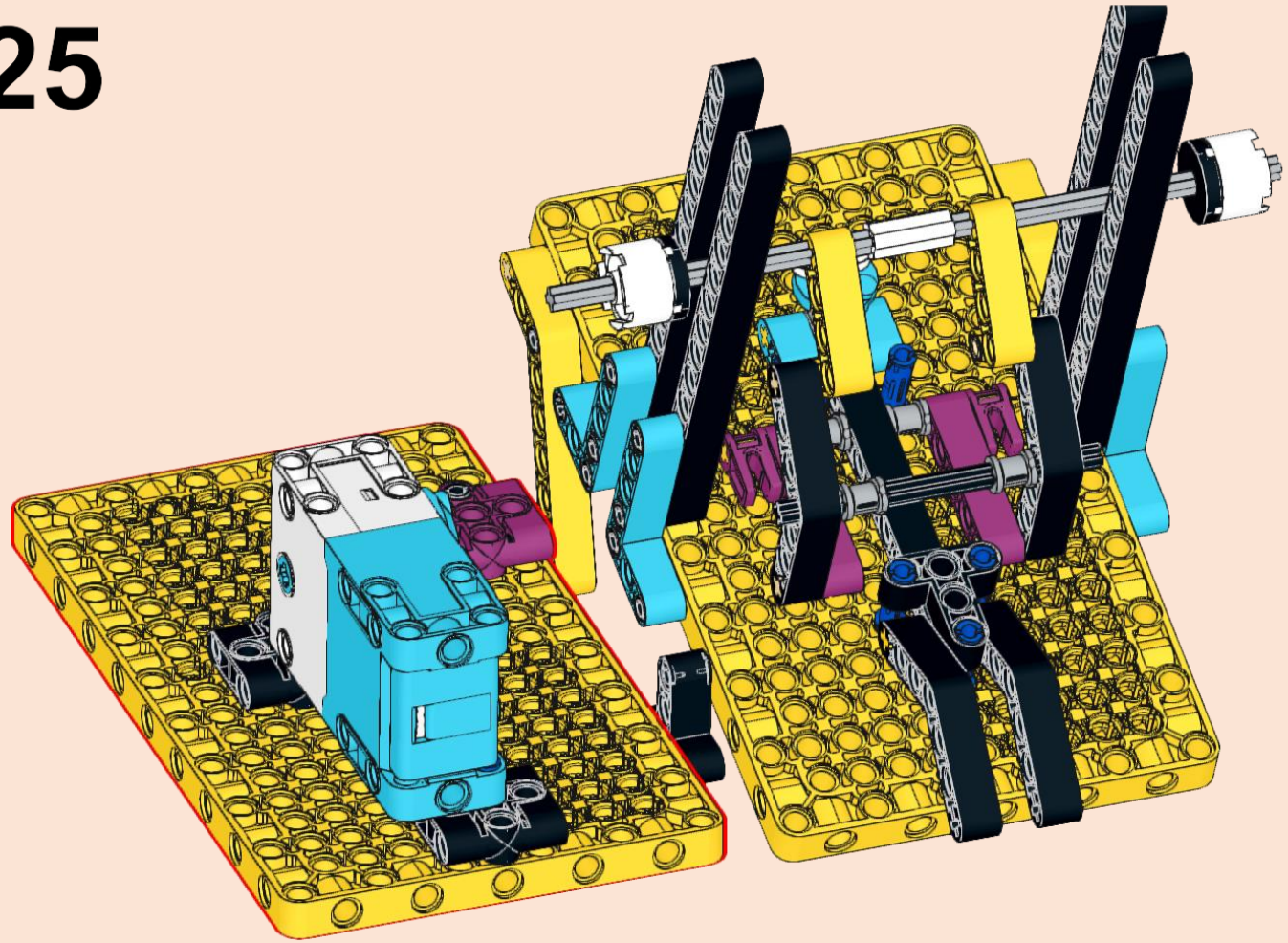




24



25



Jde se programovat!

Výzvy:

- ☐ Naprogramovat, aby motor rozhýbal činku nahoru a dolu. (použij 30% výkonu motoru)
  - Naprogramovat, aby fitnesák, pro maximální růst svalů, zdvihal činku nahoru rychle a dolu ji pokládal pomaleji.
  - + Naprogramovat jednoduché počítadlo, které na displeji ukáže počet, kolikrát fitnesák už činku zvedl.
  - + Naprogramovat, aby pravé/levé tlačítko na bricku zrychlovalo/zpomalovalo motor a rychlost cvičení

Čas na testování a úpravy



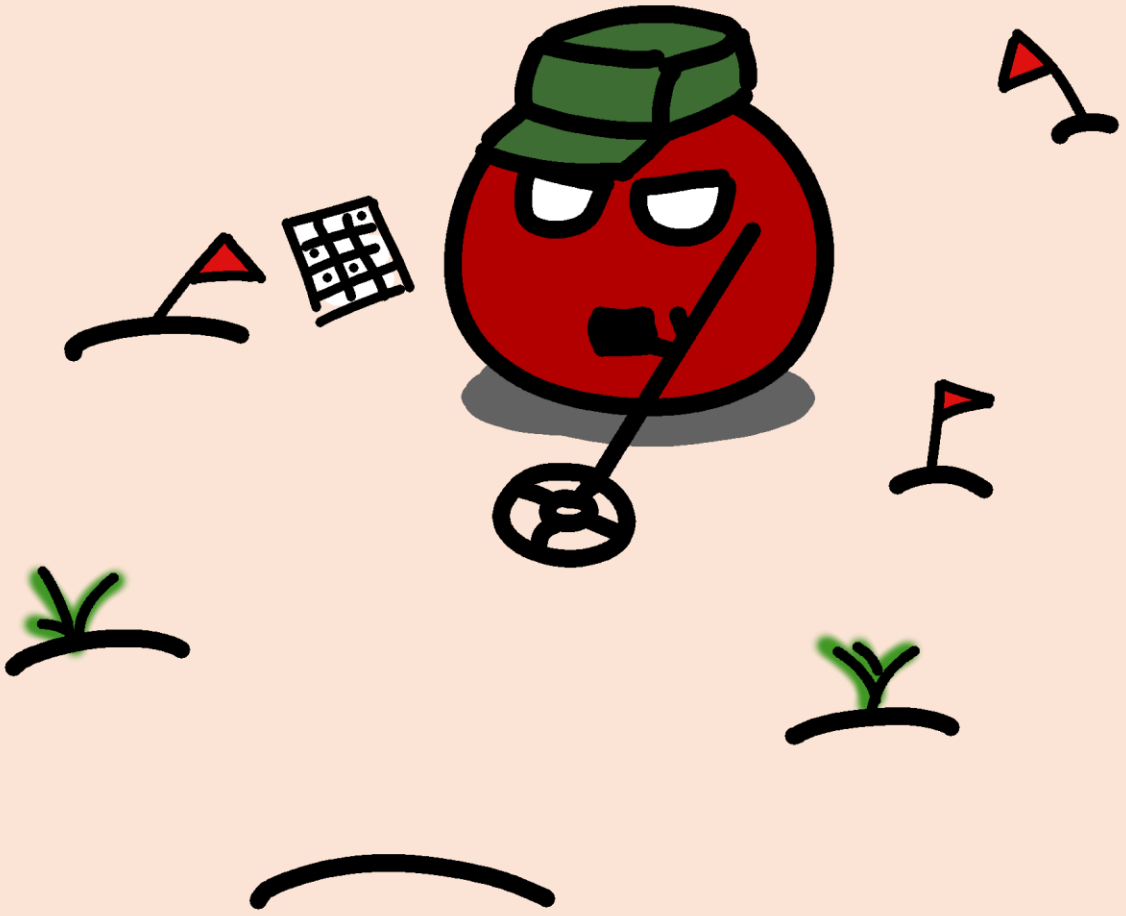
Výzvy:

- ☐ Naprogramovat časovač, který když vyprší, fitnesák přestane cvičit.
  - ○ Do samostatné proměnné bude program zaznamenávat počet zdvihů, které fitnesák dokončil.
- ☐ + Se stejnou nastavenou rychlostí motoru zkusit přestavět pohon tak, aby se činka zvedala rychleji. (převody, délka ramena...)
  - ○ + Uspořádat soutěž, čí fitnesák za 15 sekund stihne nejvíce zdvihů.

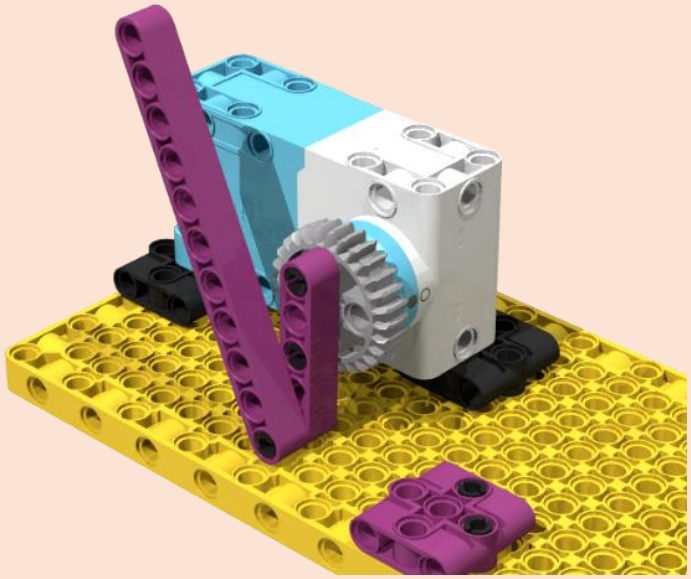
Shrnutí a zakončení

- ☐ Porovnejte s ostatníma, kdo vymyslel nejlepší program? Co ti šlo? Co ti přišlo těžký?

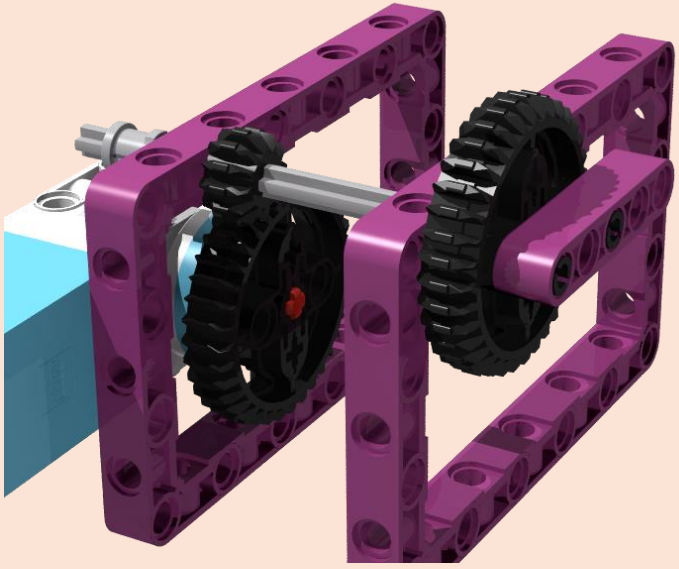




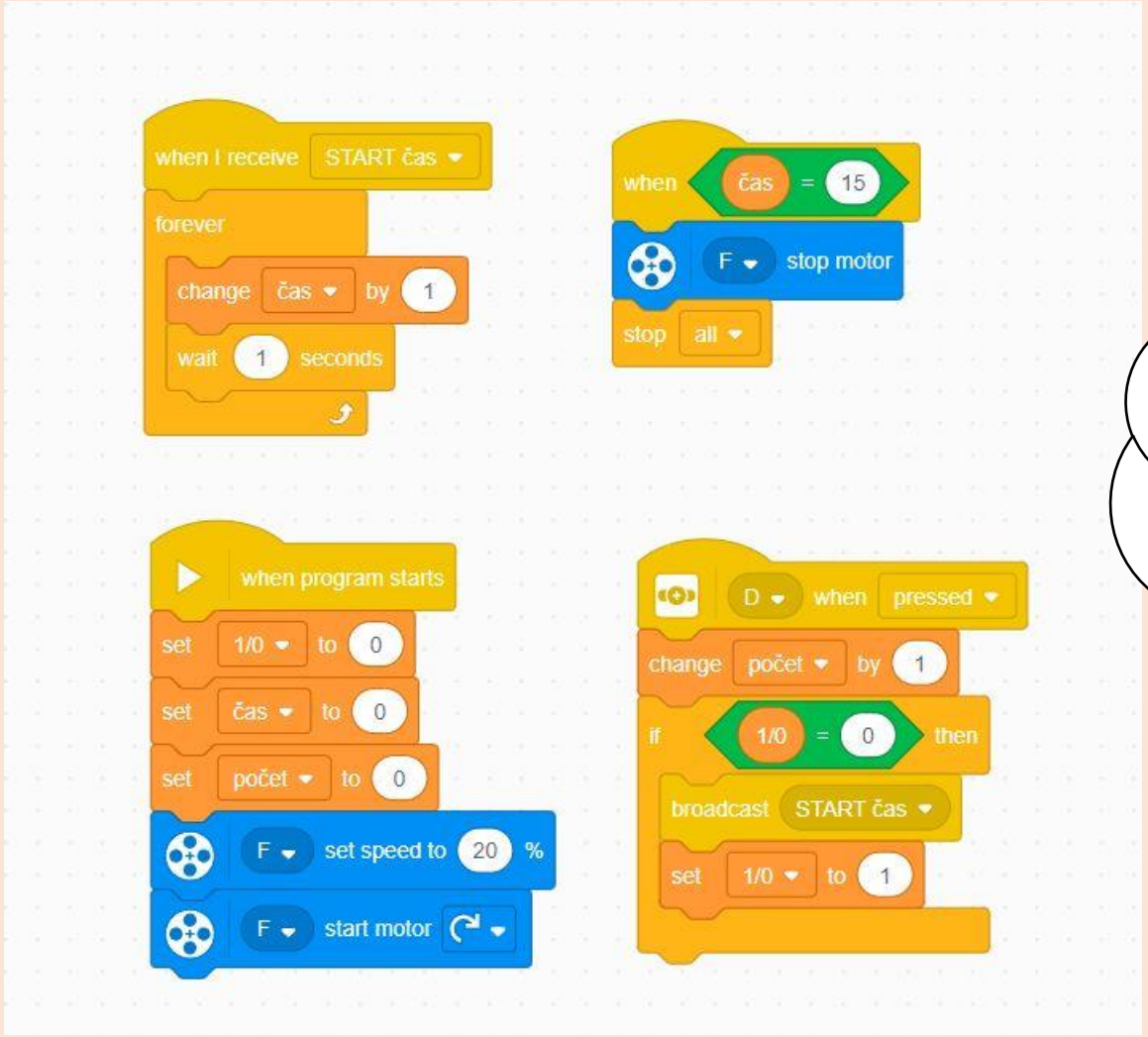
2- Posilovna



Příklad ramena pohonu



Nápad na převod pro pohon



Návrh fungujícího kódu se stopkami a počítadlem

